

Graphical analysis

Questions to ask

Why use pictures to communicate data?

What are the standard ways to use pictures to communicate data?

What are the worst ways?

What are the best ways?

How should we make sense of graphs?

Numeric systems

I, II, III, IIII ...

X, X, X, X ...

α , β , γ , δ ...

1, 2, 3, 4 ...

Roman numerals, binary, hexadecimal etc.

Displaying data

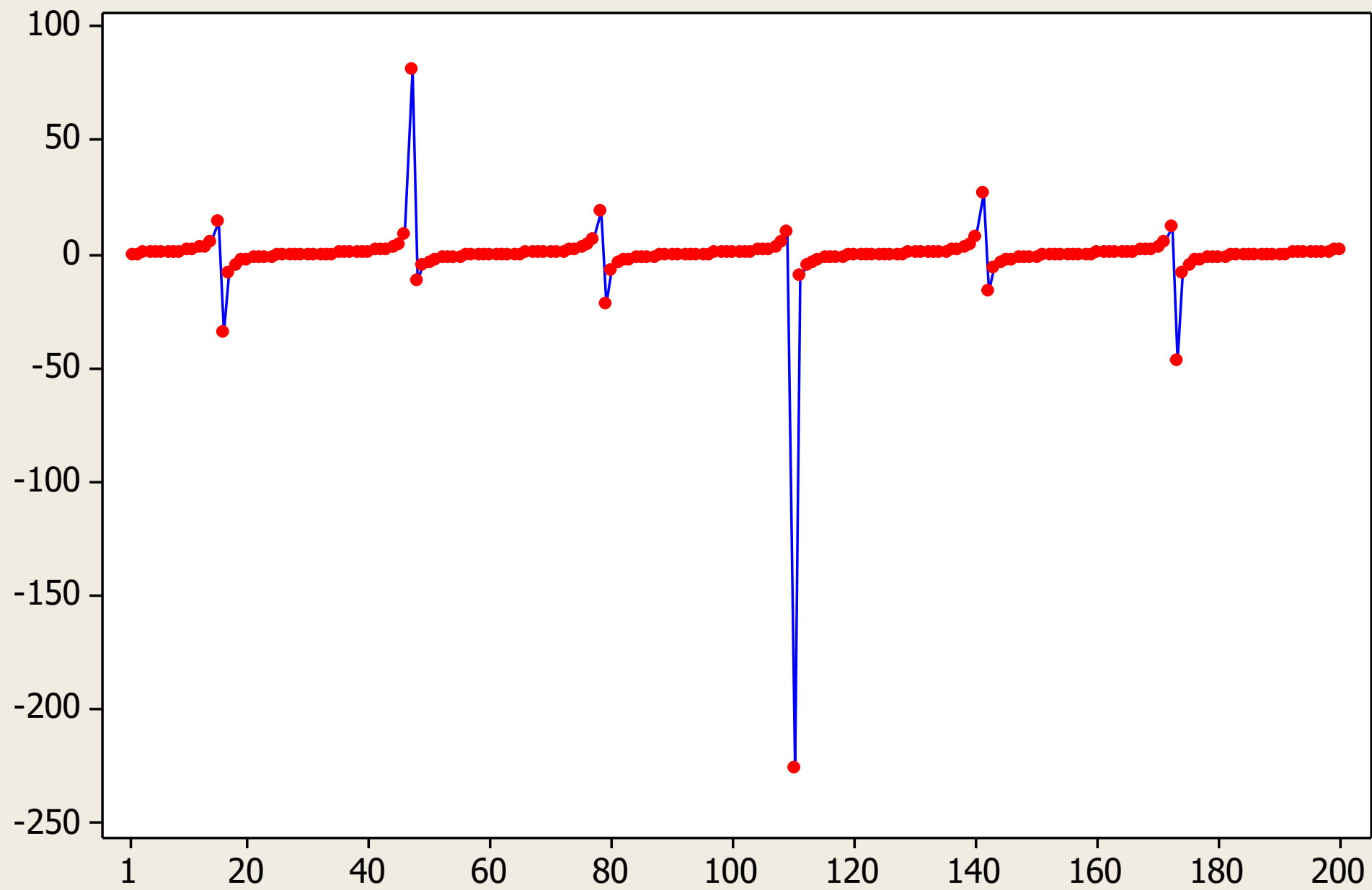
1, 2, 3, 4

Simple data set – we can ‘hold it in our heads’ easily

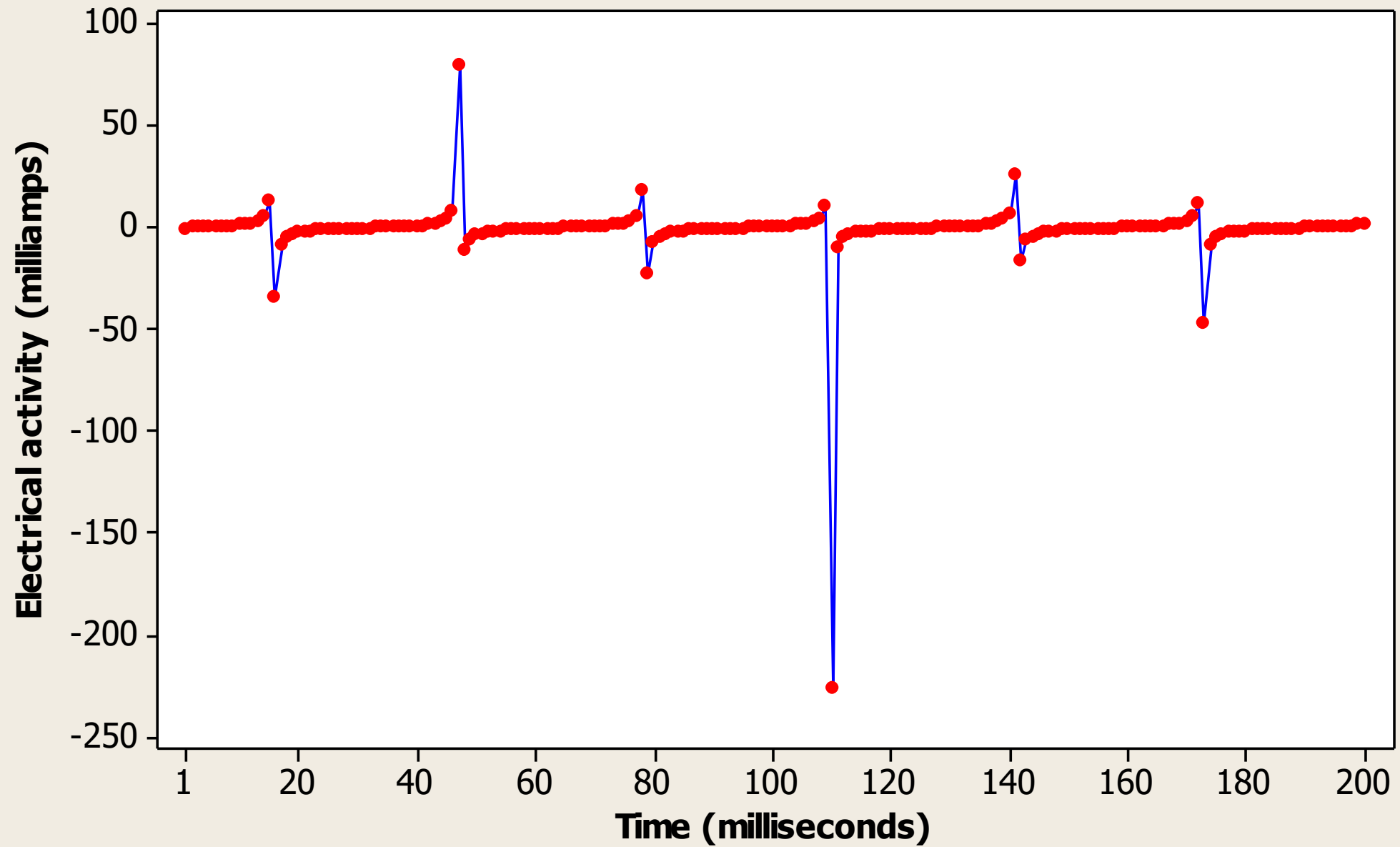
Research suggests we can only remember limited numbers of items:

Four – seven things

Nesting – chunking etc.



Cardiovascular activity



Basic charts/graphs

Difference between Chart & Graph?

Bar Chart

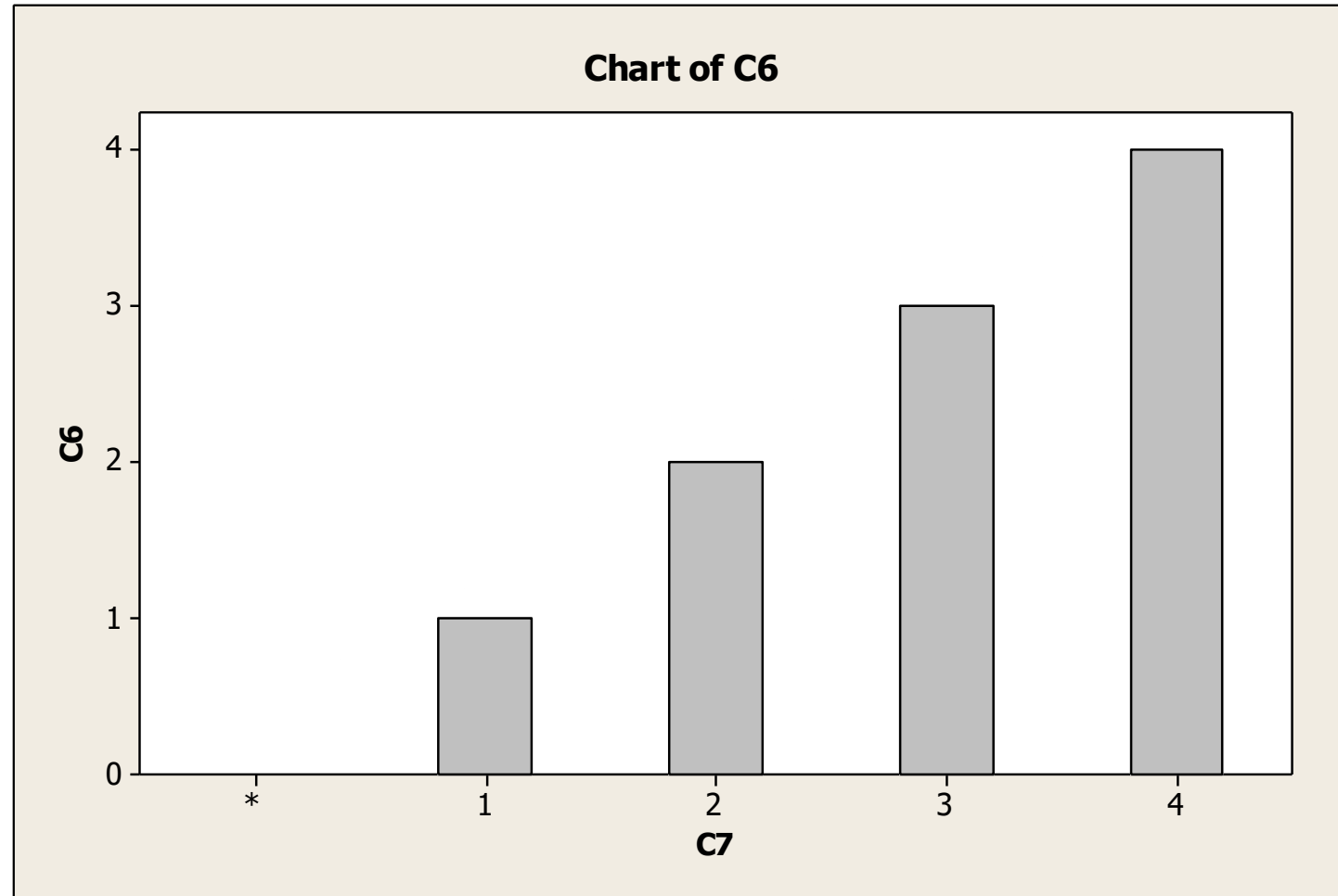
Histogram

Scatter plot

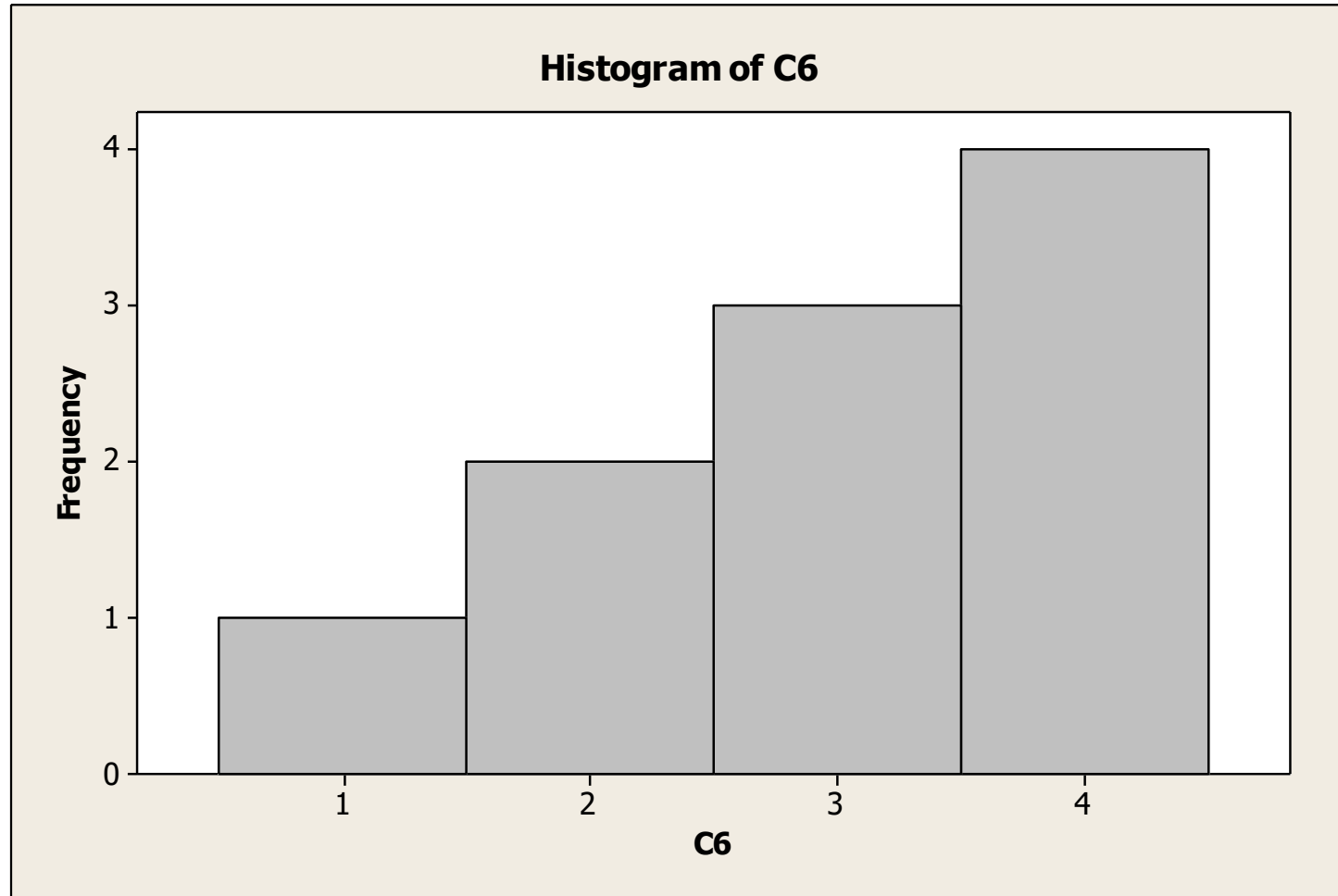
Pie chart

Dotplot \ Stem & Leaf

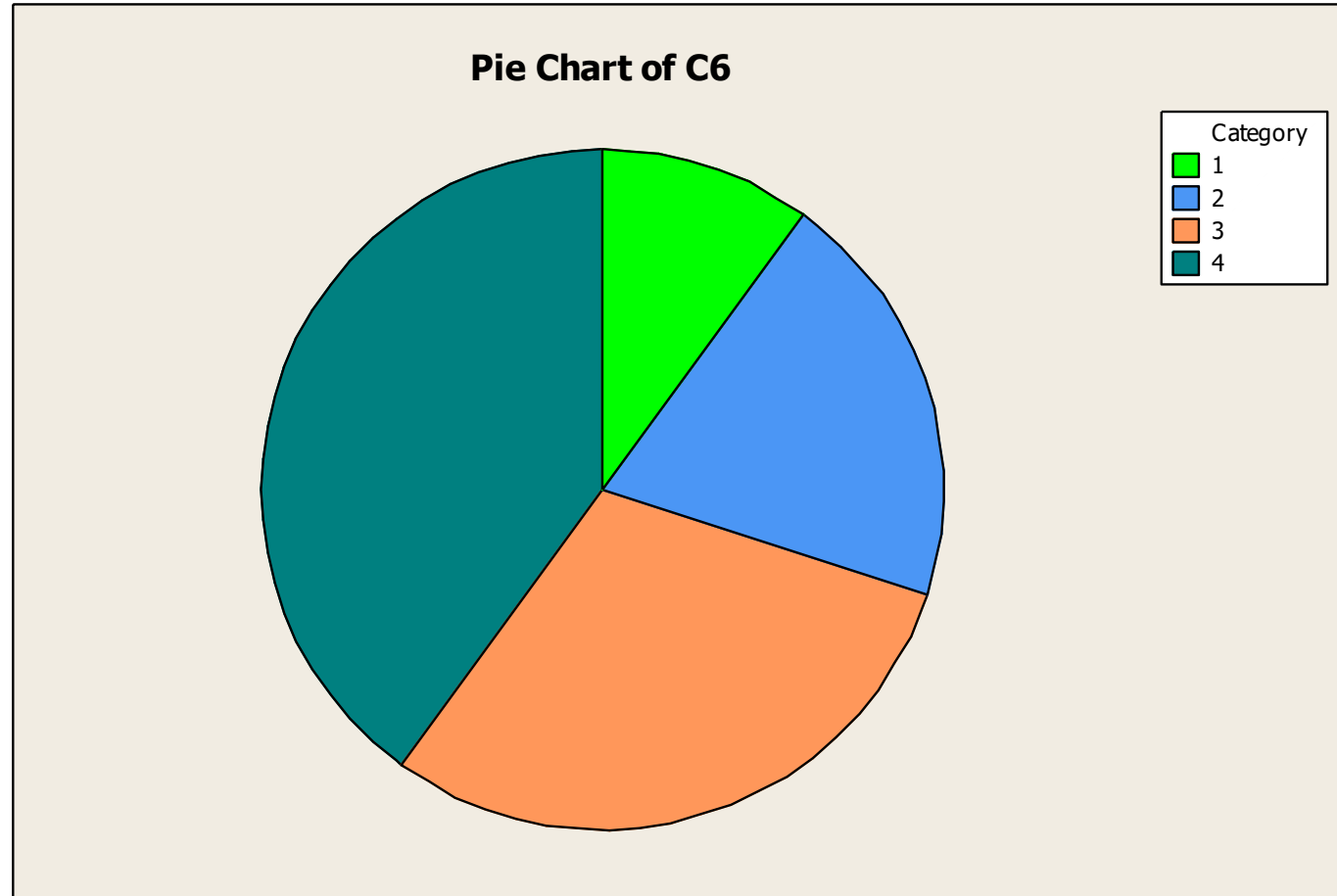
Bar



Histogram

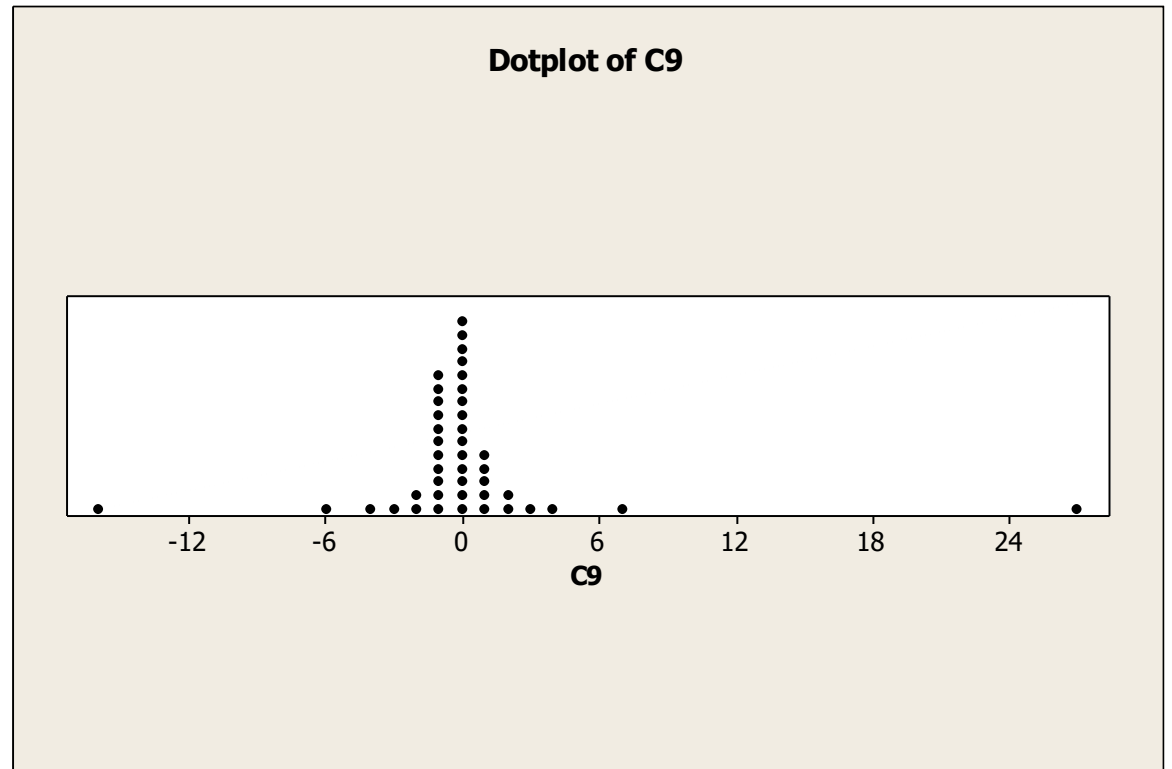


Pie



Dotplot

-1.4479 -1.17663 -0.96265 -
0.78636 -0.63586 -0.50341 -
0.38369 -0.27285 -0.16792 -
0.06647 0.033642 0.134431
0.237975 0.346585 0.463021
0.590803 0.734688 0.901475
1.101433 1.351078 1.679021
2.139842 2.85264 4.137086
7.244607 26.89324 -15.8942
-6.08689 -3.71668 -2.63407 -
2.00408 -1.58503 -1.28098 -
1.04618 -0.85599 -0.69589 -
0.55669 -0.43221 -0.31808 -
0.21101 -0.10838 -0.00796
0.092297



Stem & leaf

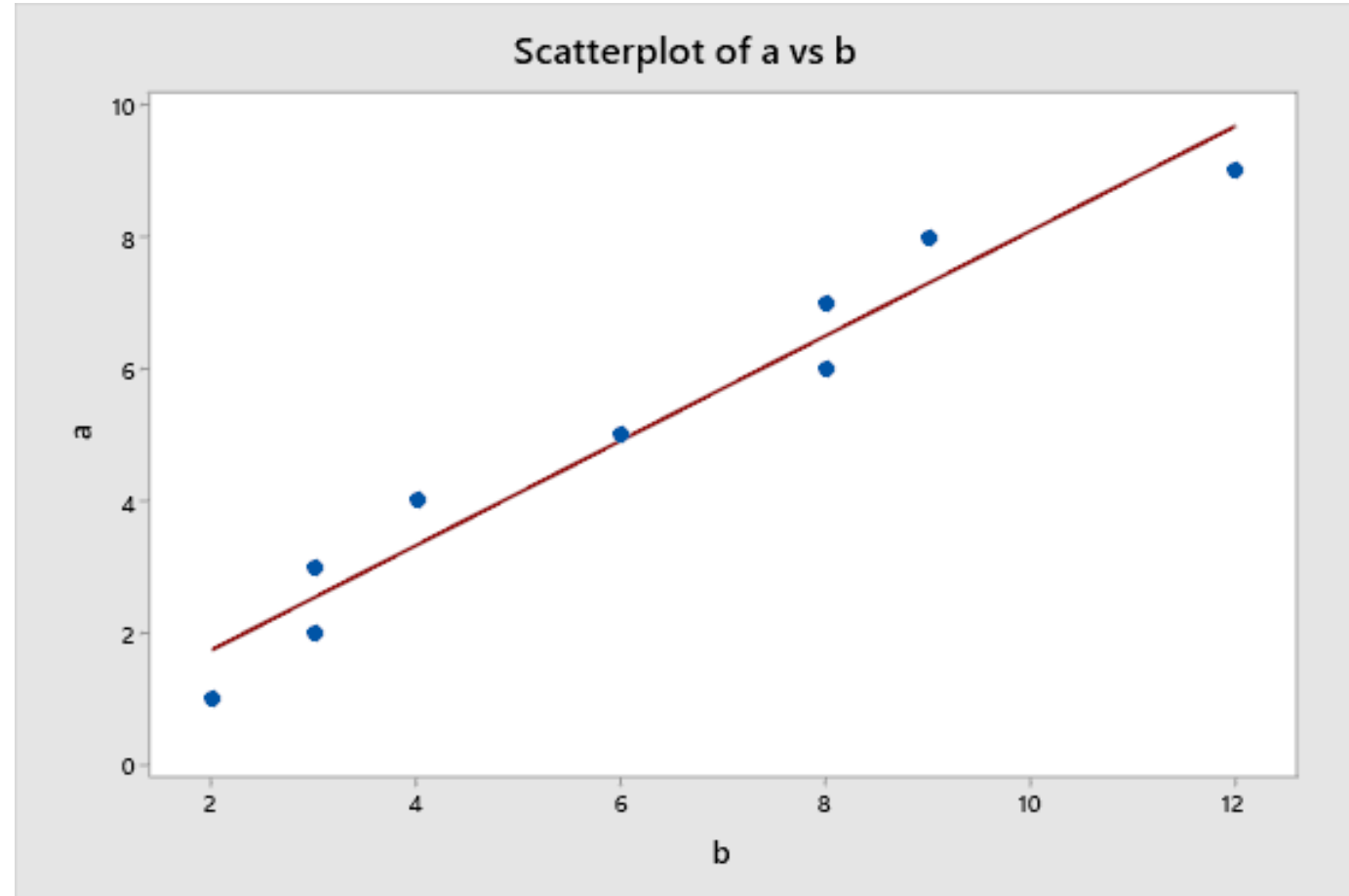
1	-1	5
0	-1	
2	-0	6
(24)	-0	322111110000000000000000
17	0	000000000111224
2	0	7
0	1	
0	1	
1	2	
1	2	6

Scatterplot

Used to show the relationship between variables or measurements.

Often includes a trendline

Connect the dots **only when you have reason** (e.g. time points)



Outliers

If you think that there is something wrong with a data point you can remove it – you should always **say that** you have done so and **why** you have done so.

Just because a data point is **unusual** doesn't mean you should always remove it!!!

Consider the distribution of ages in this room...

Bad Graphs

Do not communicate information well

Could be due to:

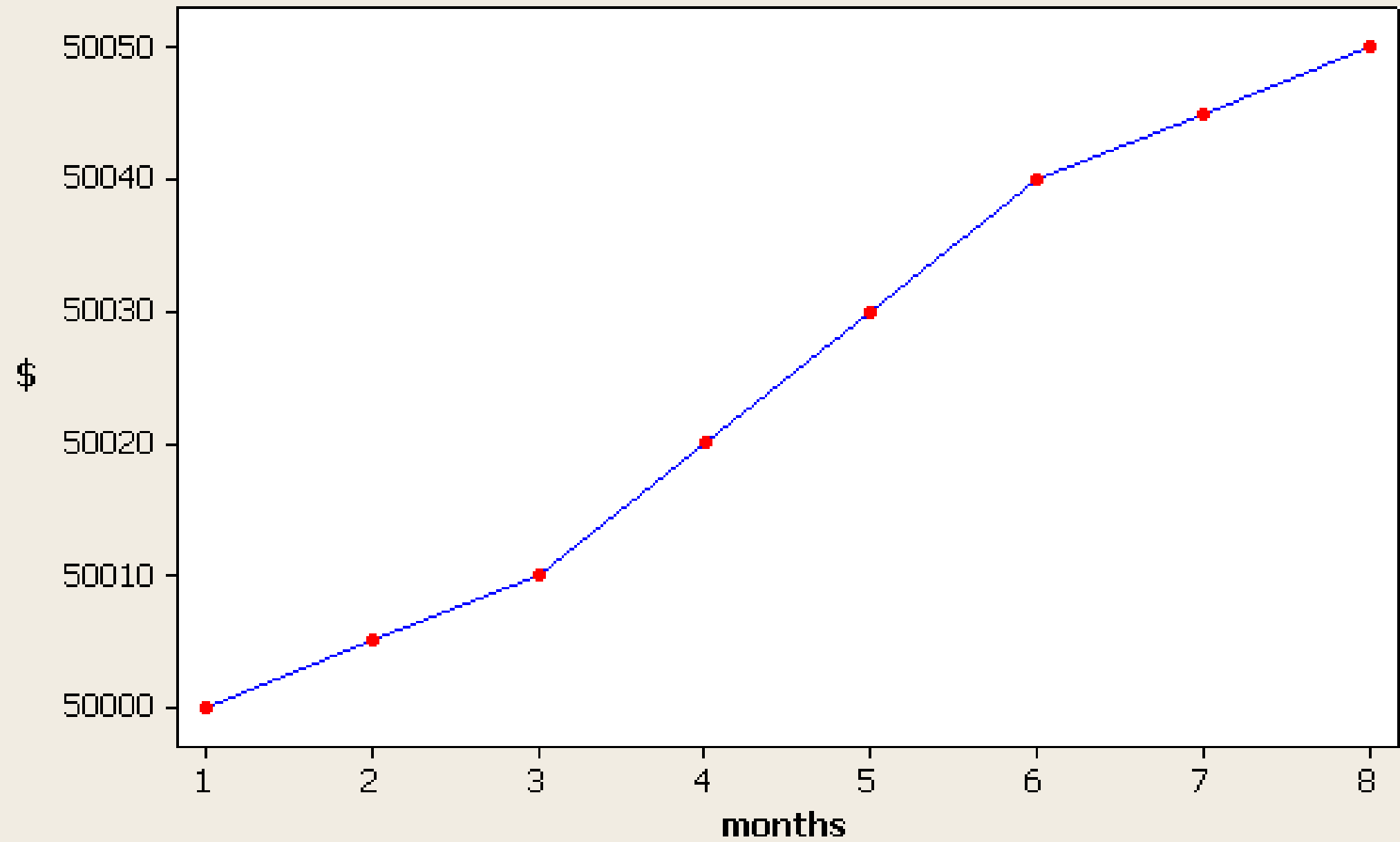
- Deliberate hiding/twisting of information

- Lack of index, title, axes labels

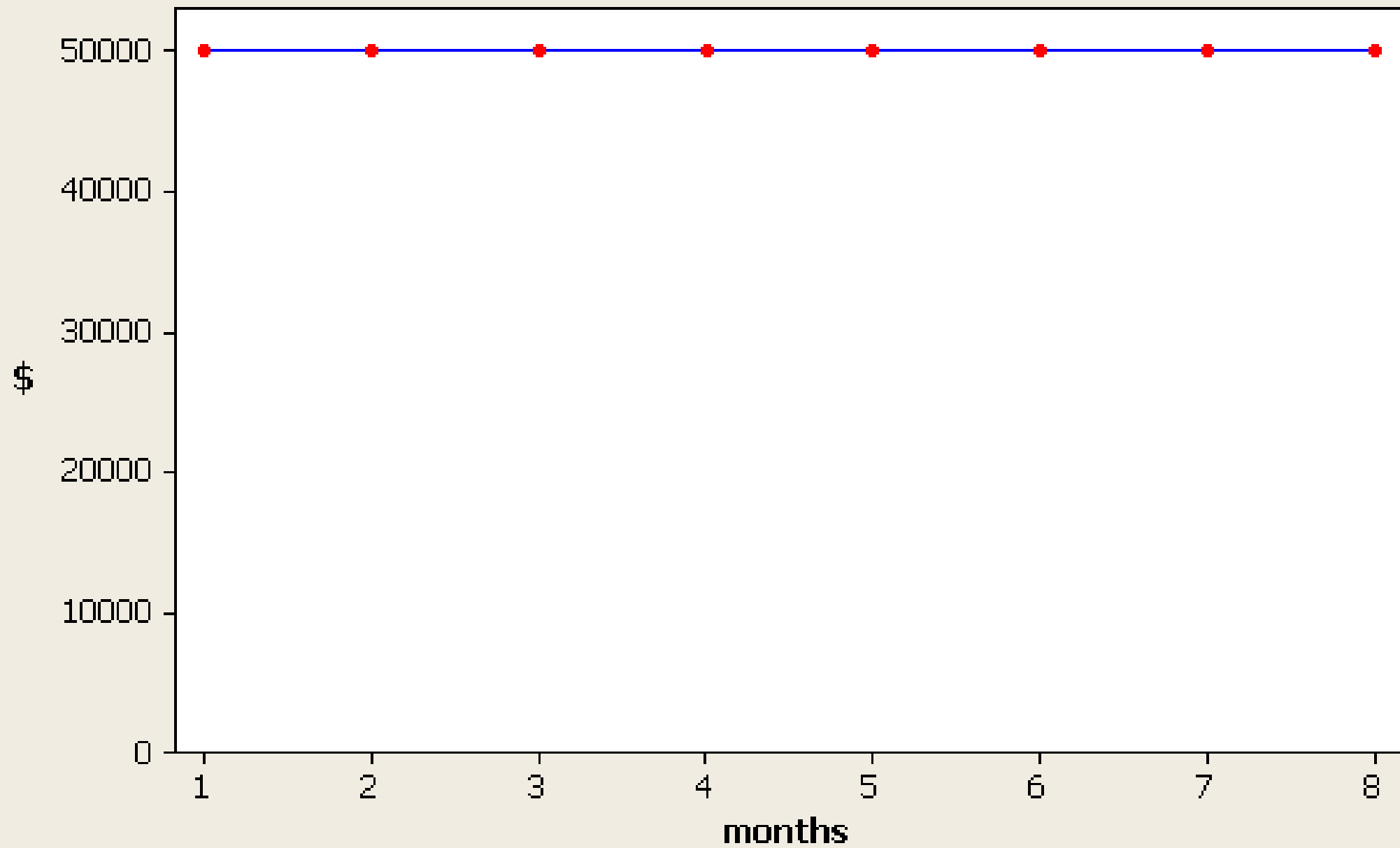
- Poor use of colour/line width/symbol use

- Poor choice of scale

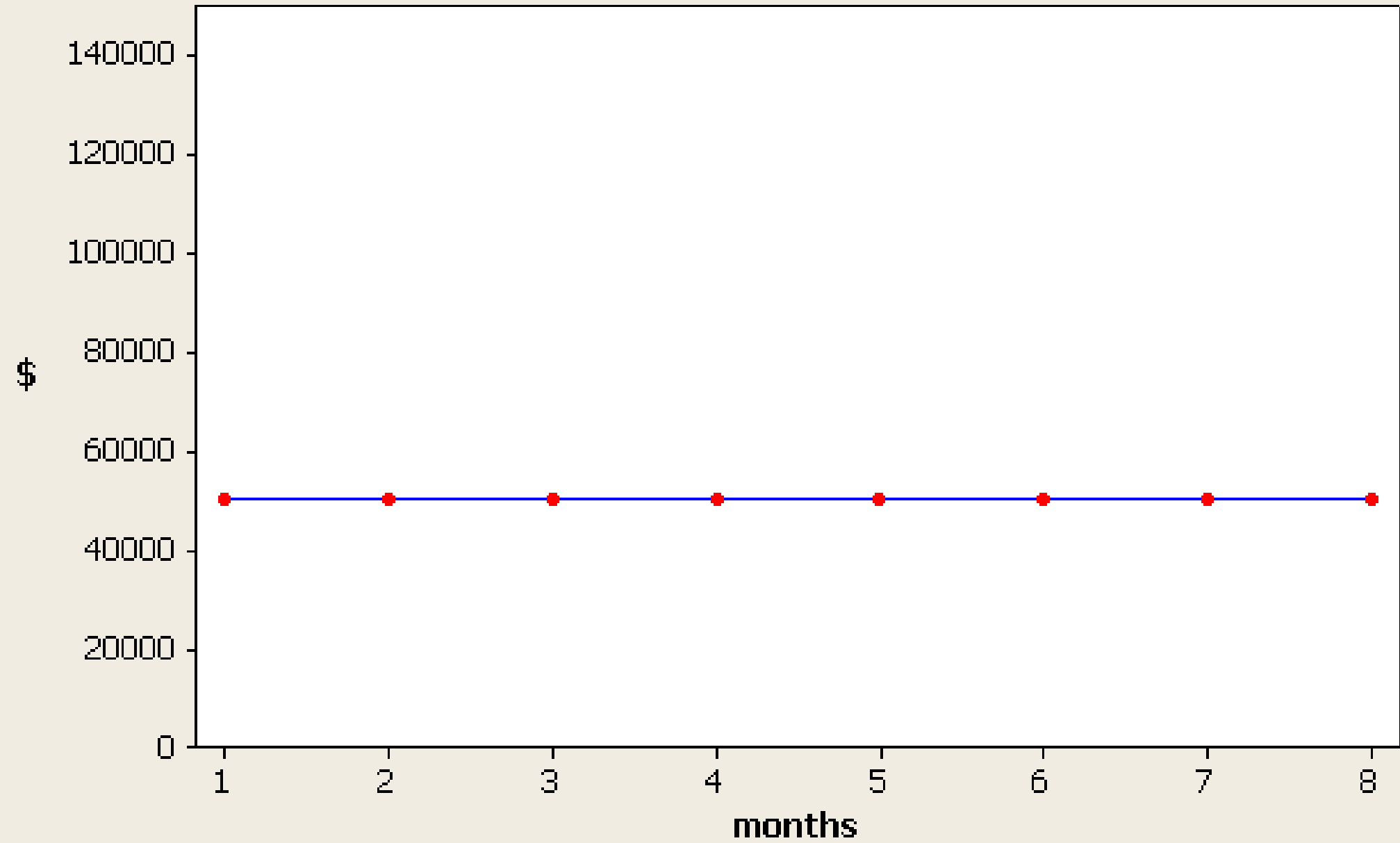
Company profit



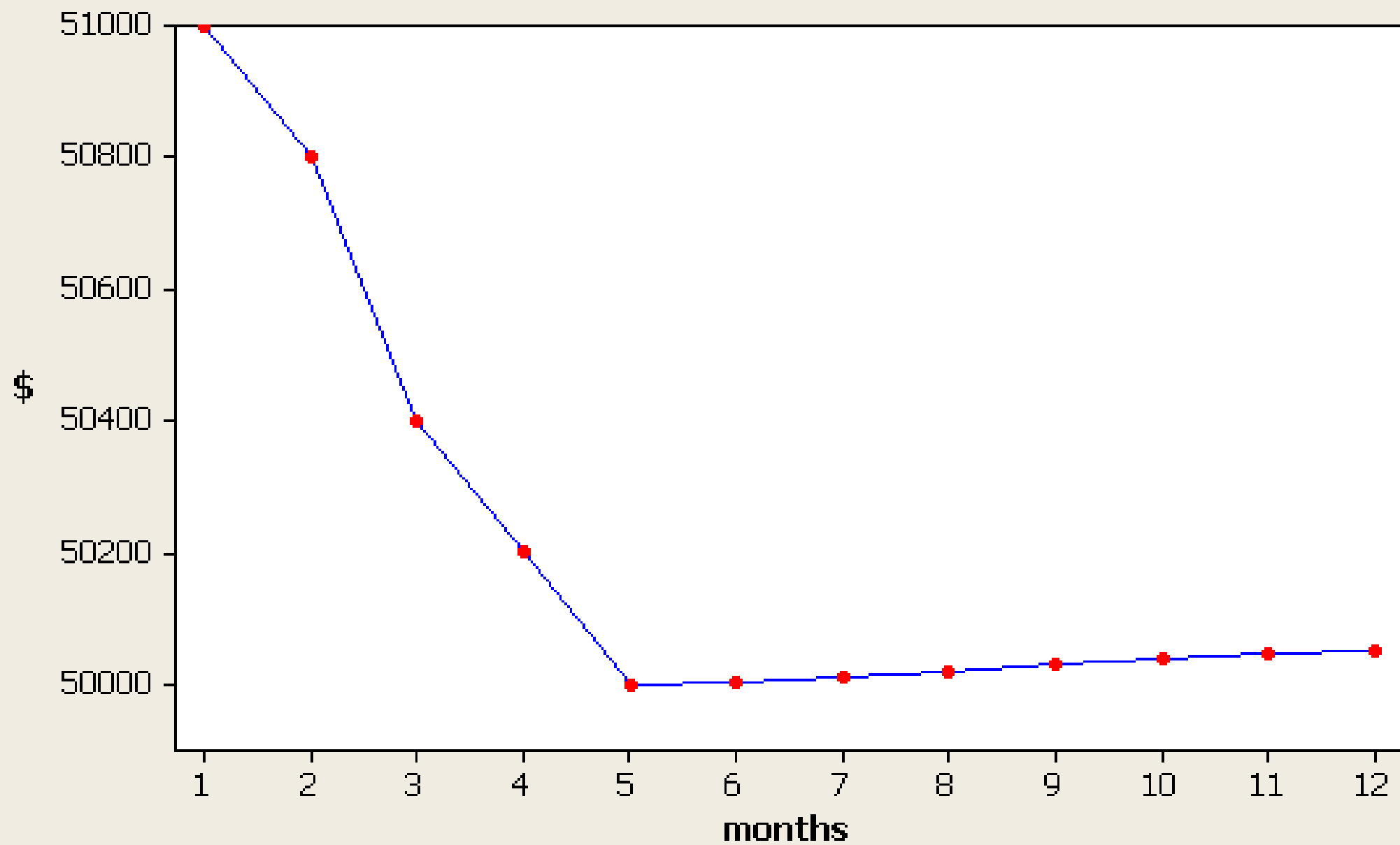
Company profit



Company profit

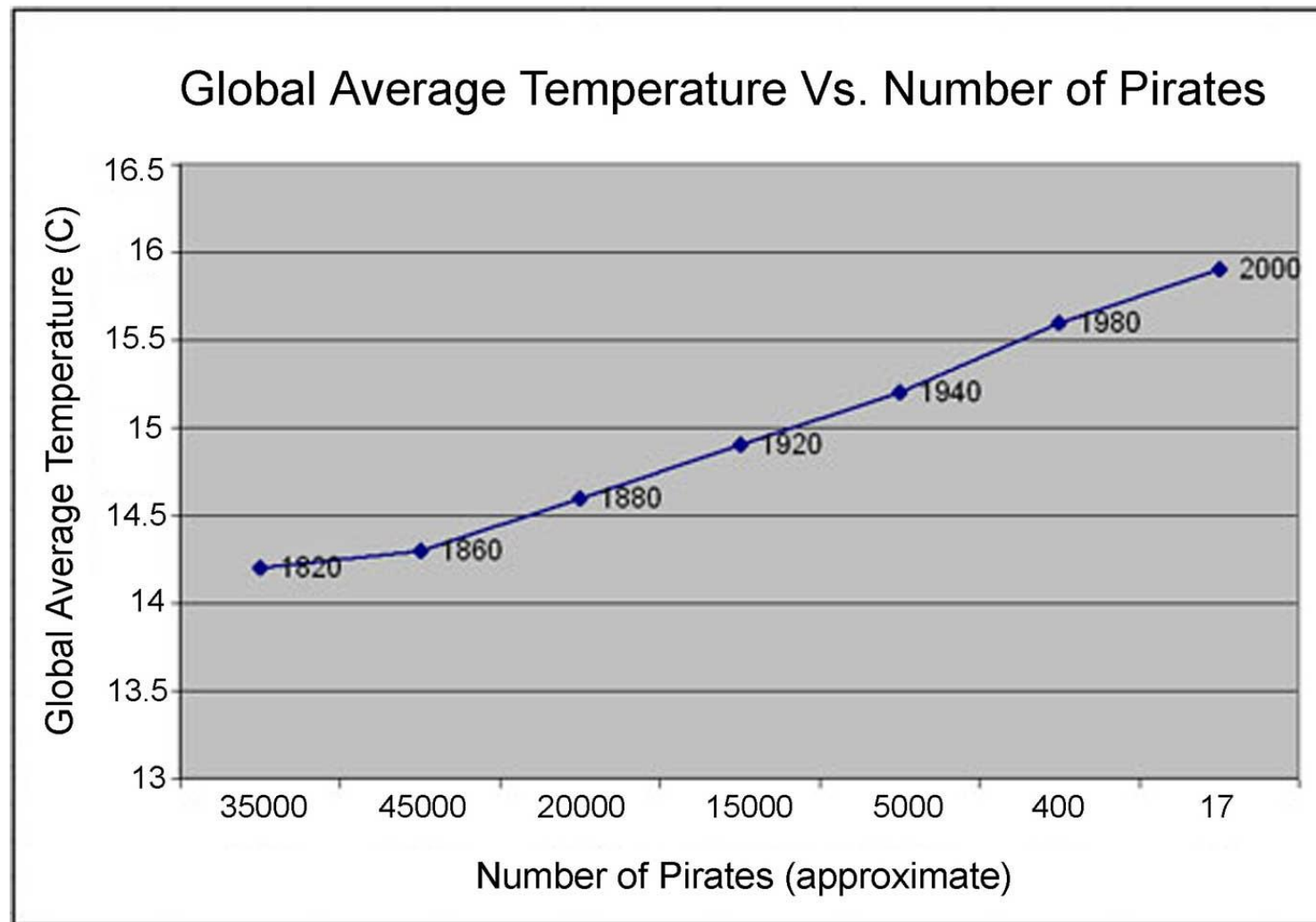


Company profit



Correlation does not equal causation!

STOP GLOBAL WARMING: BECOME A PIRATE



Causation and directionality



Good Graphs

Make the data more easily communicable

Can be understood with little effort

Can deal with complex information

Are well labelled

Are Beautiful!!!

900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	0	100	200	300	400	500
10	10	11	12	12	12	12	12	12	11	11	10	10	10	10	10	9	9	9	9	9
11	11	11	13	13	13	13	13	12	11	11	10	10	10	9	9	8	8	8	9	9
N	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	NNE	N	N	N	N	N	N
86%	85%	82%	76%	73%	70%	70%	72%	76%	78%	80%	81%	81%	82%	83%	83%	84%	84%	84%	84%	86%
G	G	G	G	G	G	G	G	G	G	G	G	G	VG	VG	G	G	G	G	G	G
1024	1024	1024	1025	1025	1024	1024	1024	1024	1025	1025	1025	1025	1025	1025	1025	1025	1024	1024	1024	1023

Y TYWYDD NOTTINGHAM

Ymhellach ymlaen ▶

+ Ychwanegu i'ch ffefrynau

⚙ Gosodiadau

laith

Gwe



13°C
9°C

13

Sad



12°C
9°C

13

Sul



12°C
9°C

8

Llu



12°C
7°C

5

Maw



12°C
7°C

9

Dod o hyd i Ragolwg

Rhowch dref, sir neu gôd post y DU yma



> Hoff Leoliadau (1)

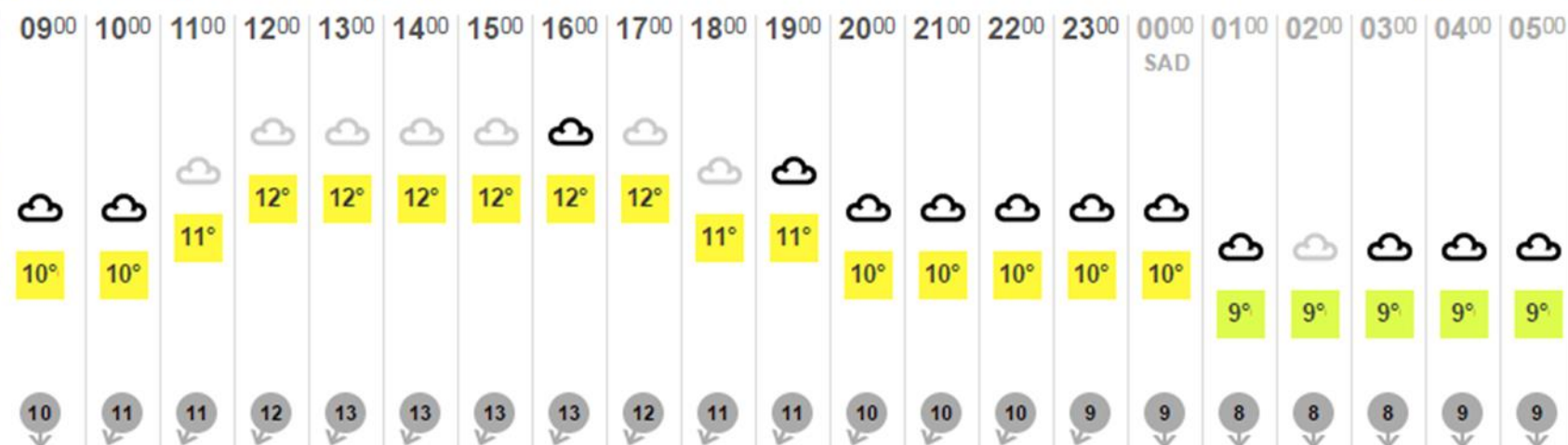


Codiad haul 07:32 Machlud haul 18:08

Diweddarwyd 08:28

Graff

Tabl



East Midlands

Rhagolwg Cryno

Mae'r cynnwys hwn ar gael yn Saesneg yn unig.

Rhagolwg Amgylcheddol

UV: 2

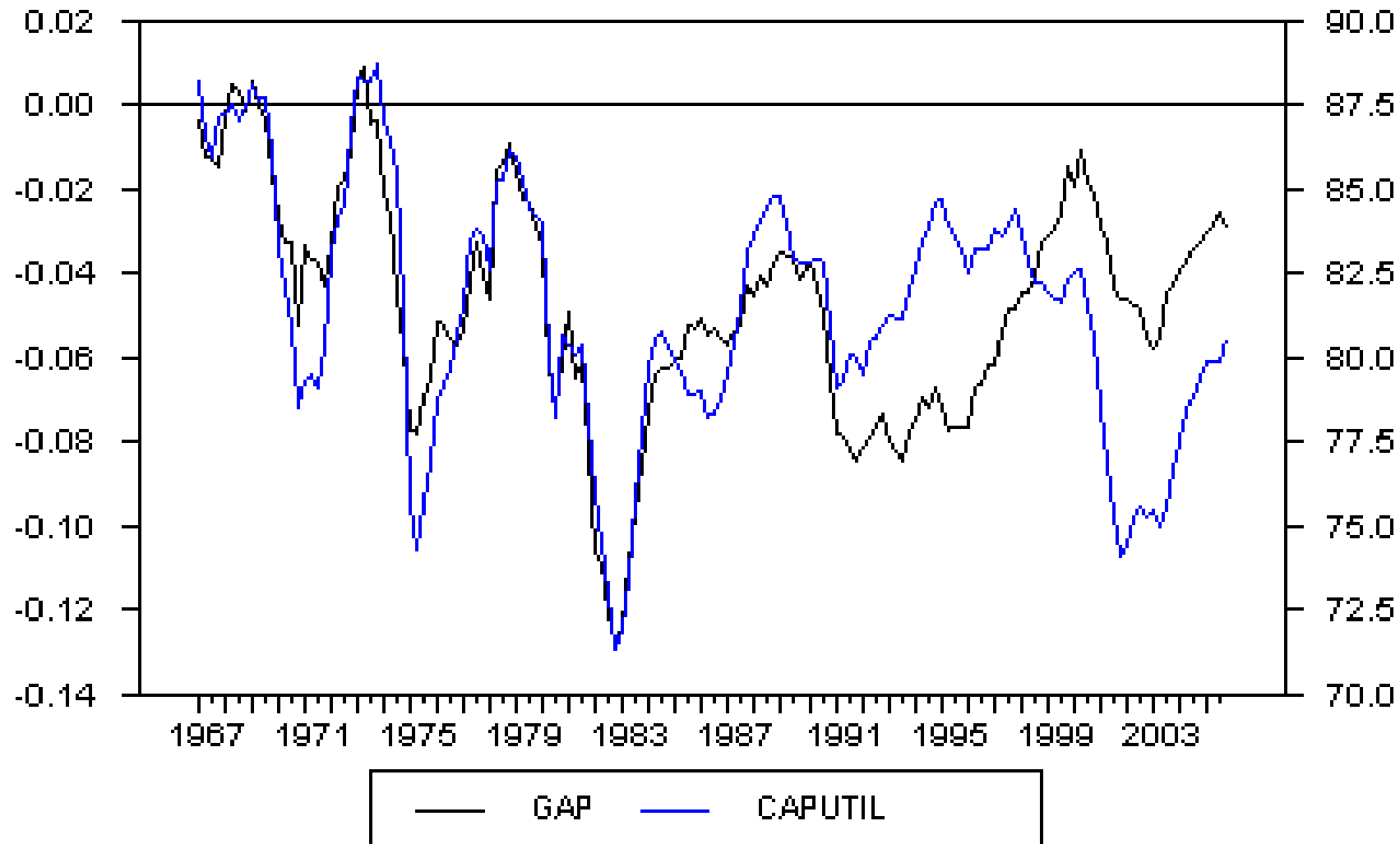
Fideo Rhagolwg

BBC East Midlands Today DU

Share



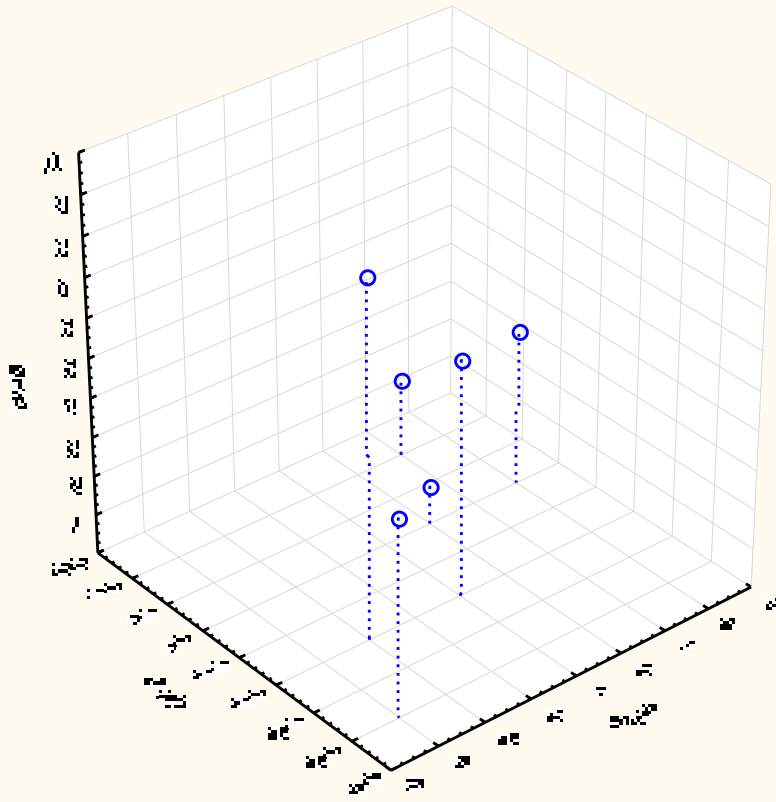
Advanced Graphs: dual axes



Advanced Graphs: triple axes

3D Scatterplot of Var3 against Var1 and Var2

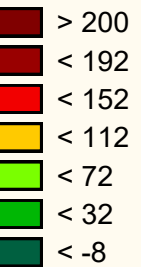
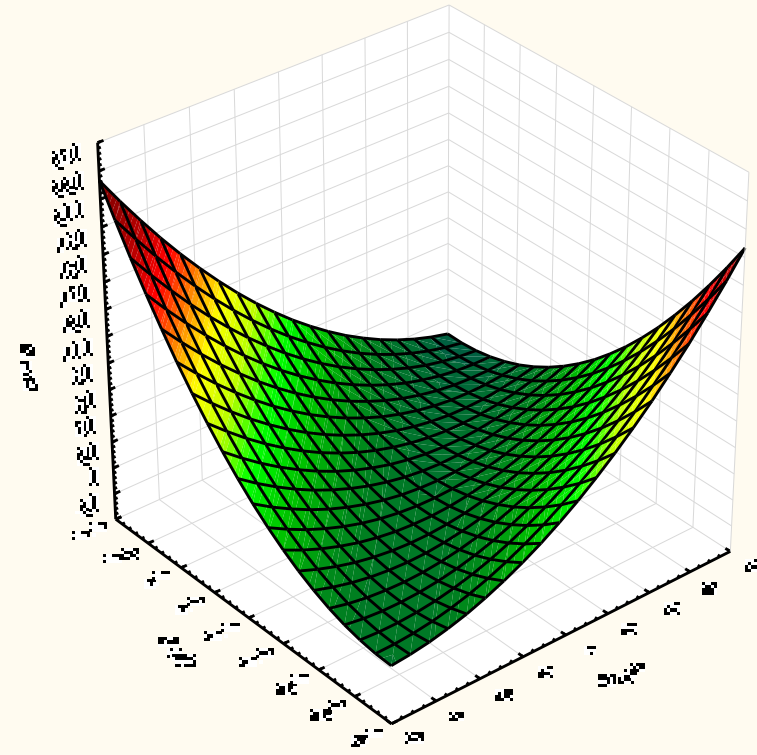
Spreadsheet1 10v*10c



3D Surface Plot of Var3 against Var1 and Var2

Spreadsheet1 10v*10c

Var3 = Distance Weighted Least Squares



Advanced graphs: multiple axes

Self-reported well-being

Population thriving by category, % of total, 2013

- World
- Brazil
- India
- Russia
- China
- South Africa

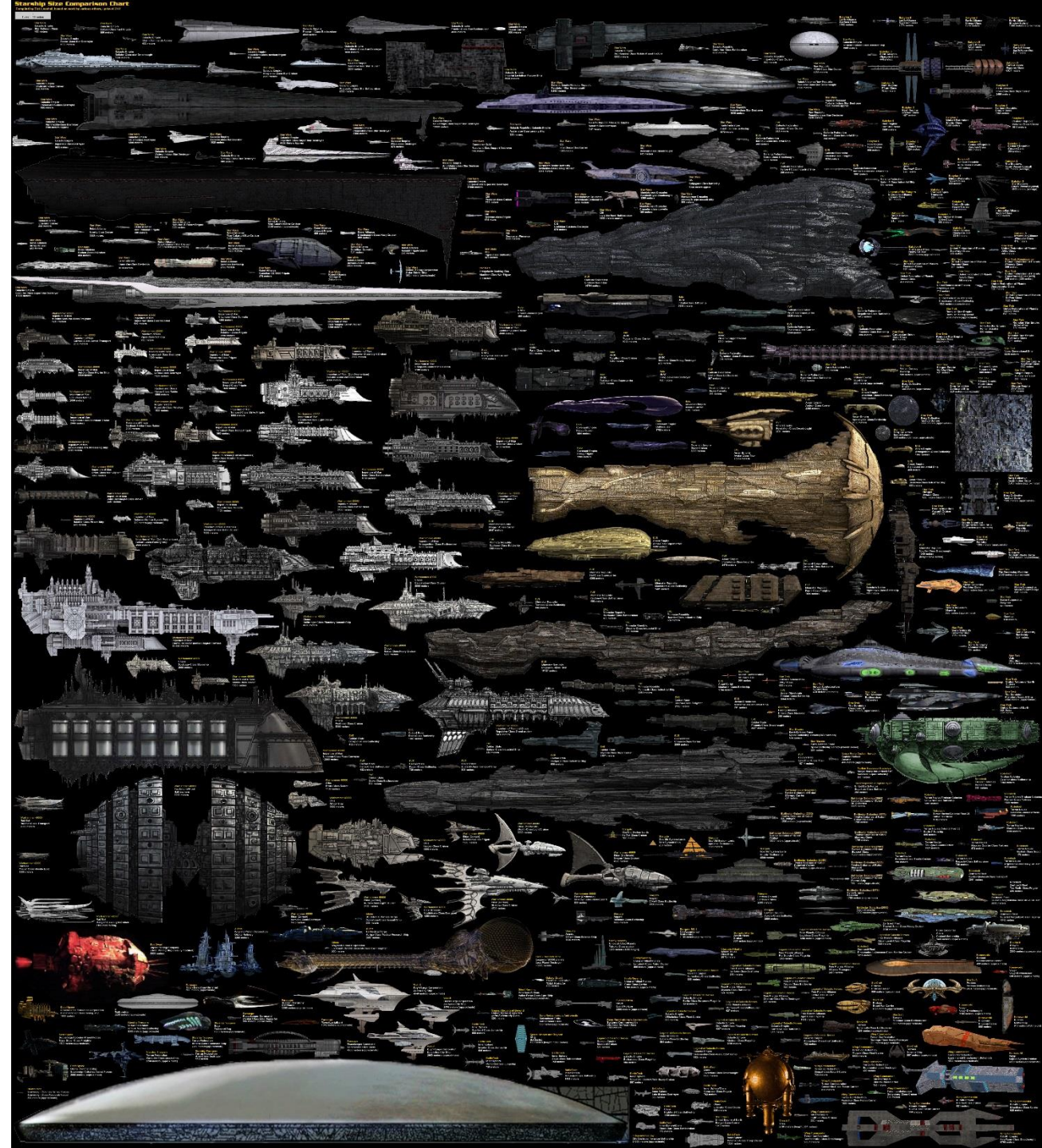


Infographics

Very modern

Very useful

Very easy to lie with!

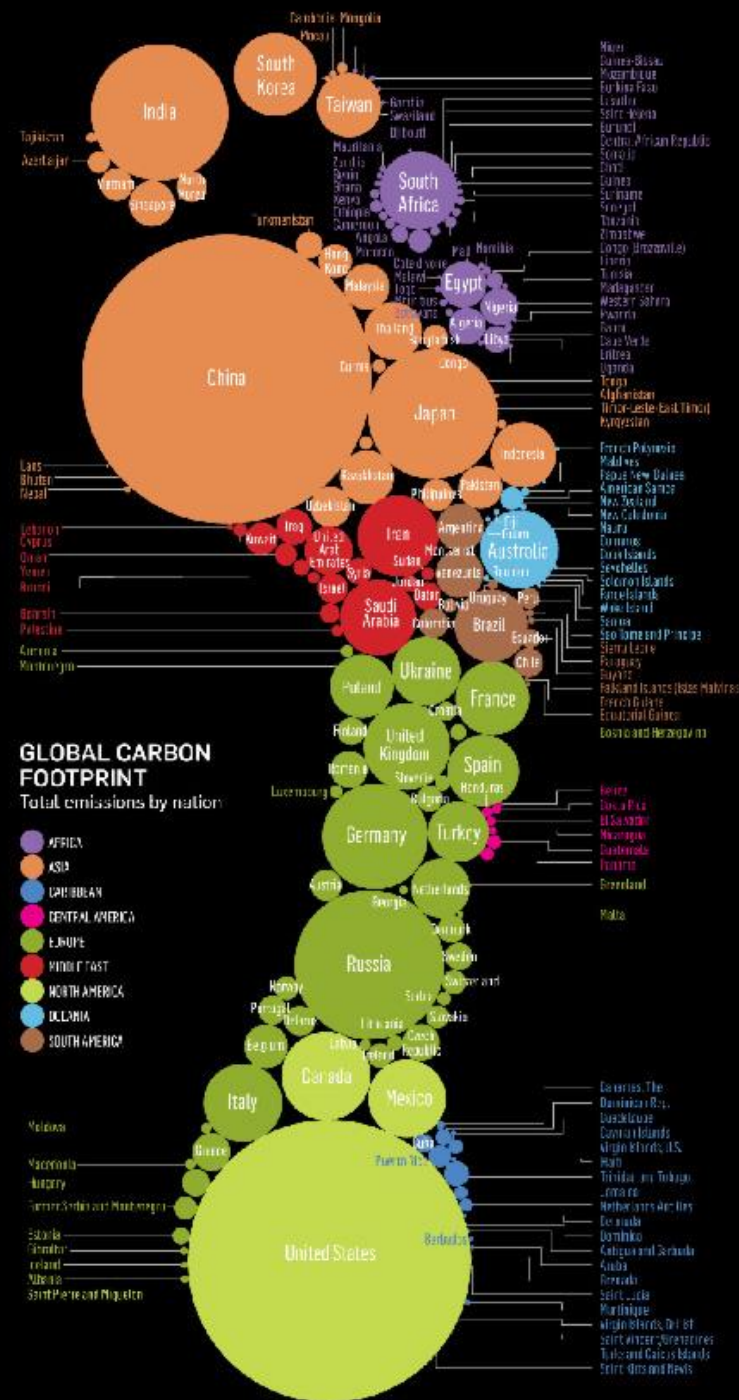


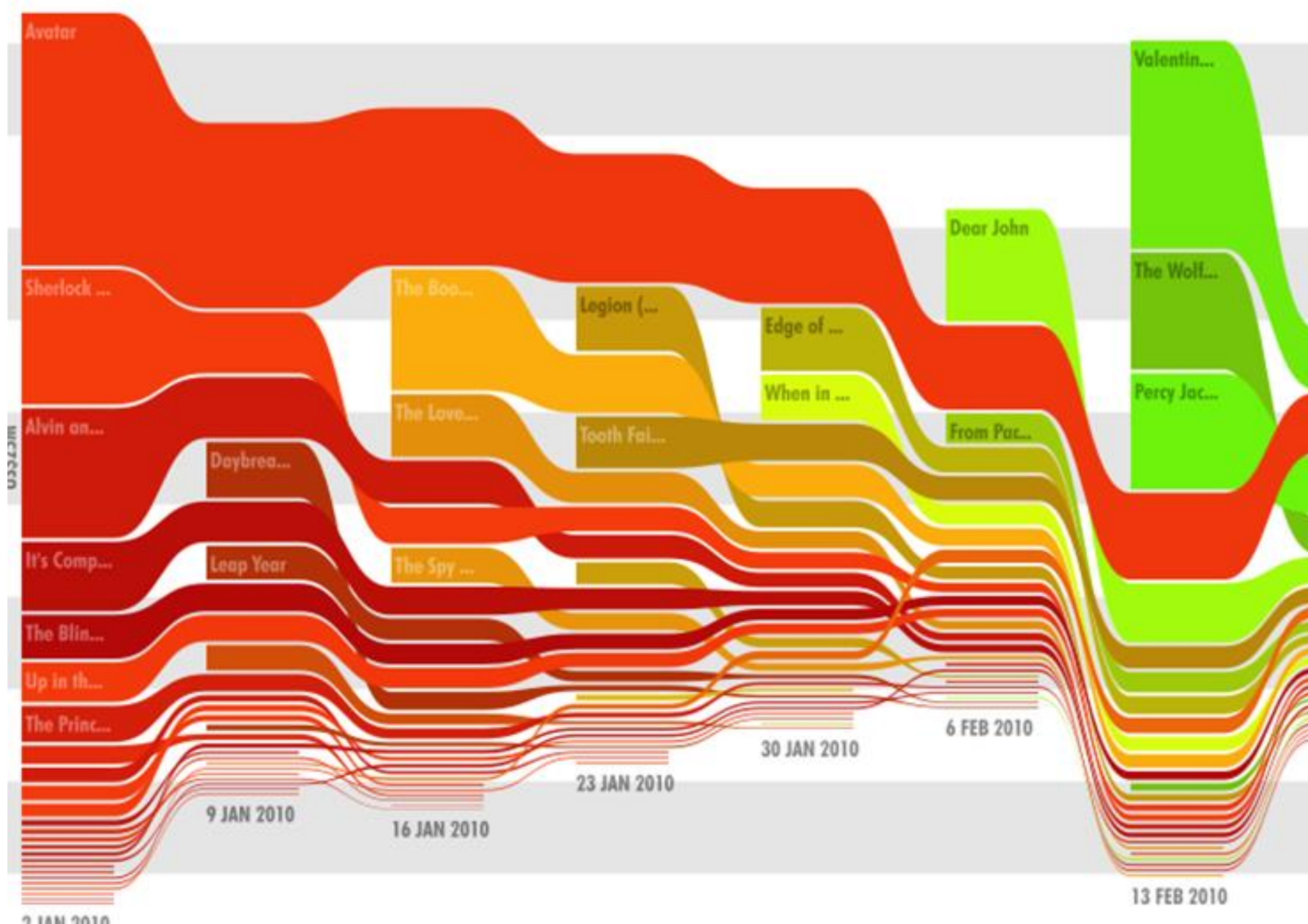
The NSW Health system is...

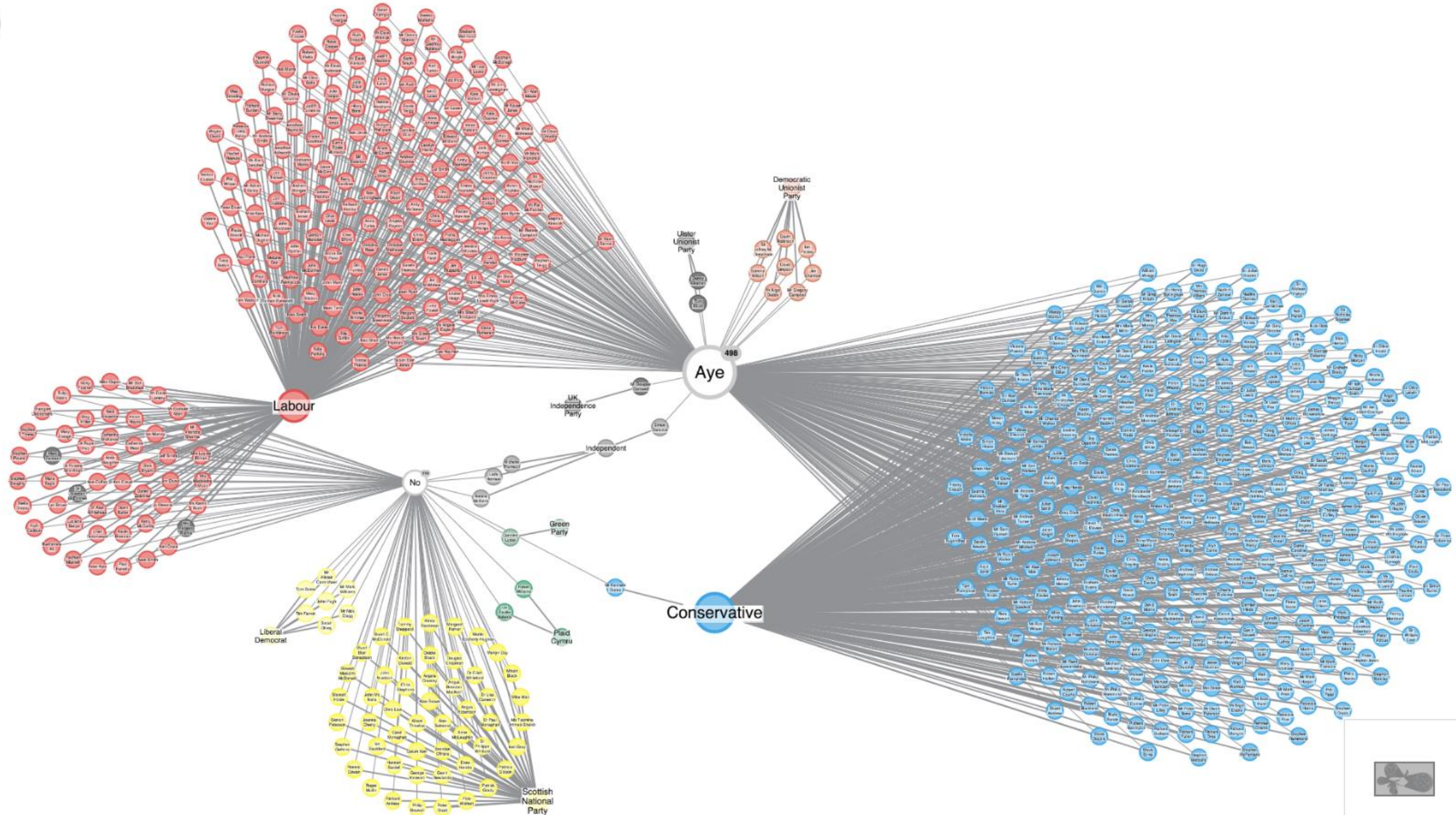
▶ RECRUITING MORE NURSES*



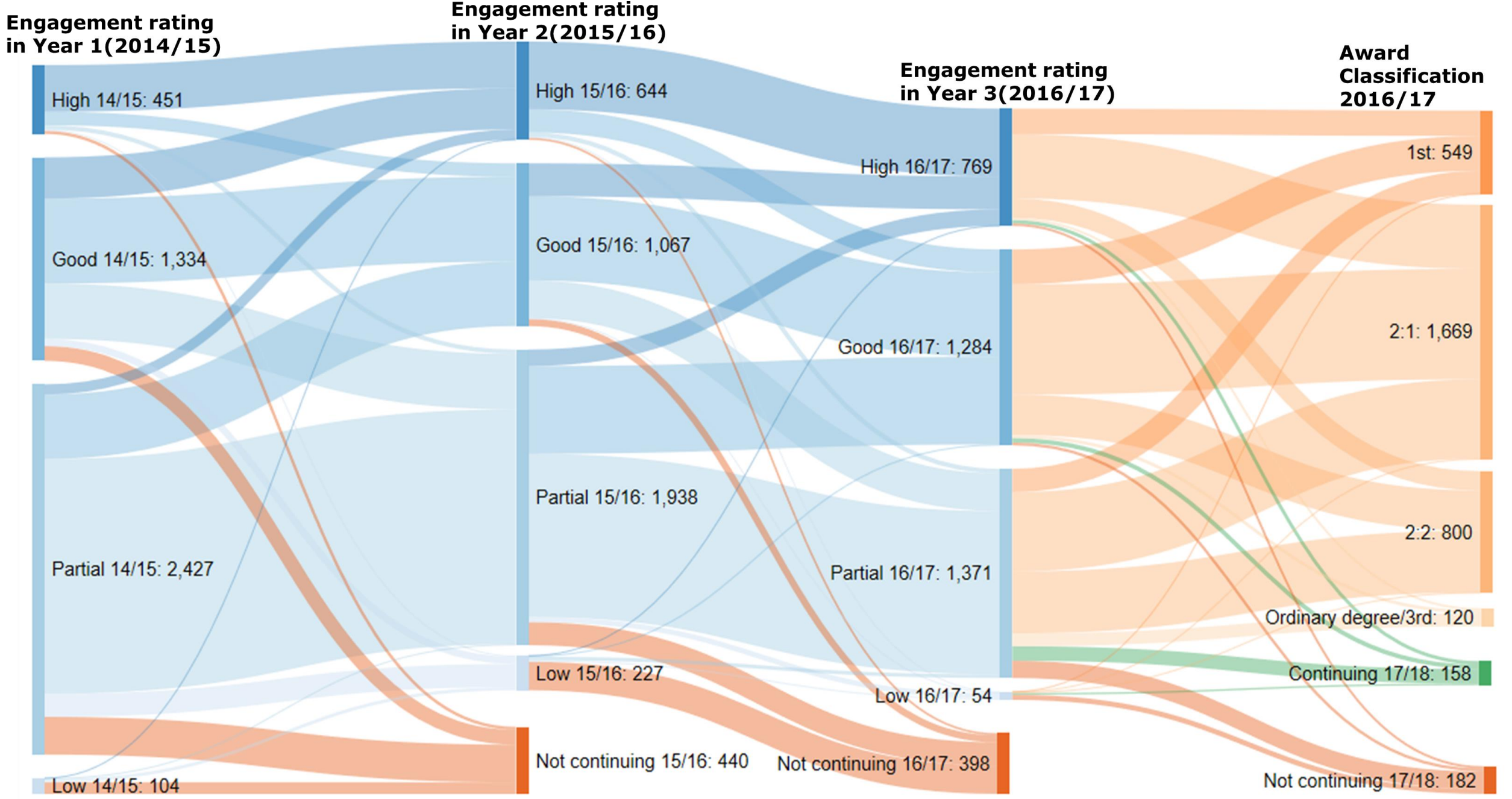
* Nursing headcount figures at June includes non casual staff and 3rd schedule



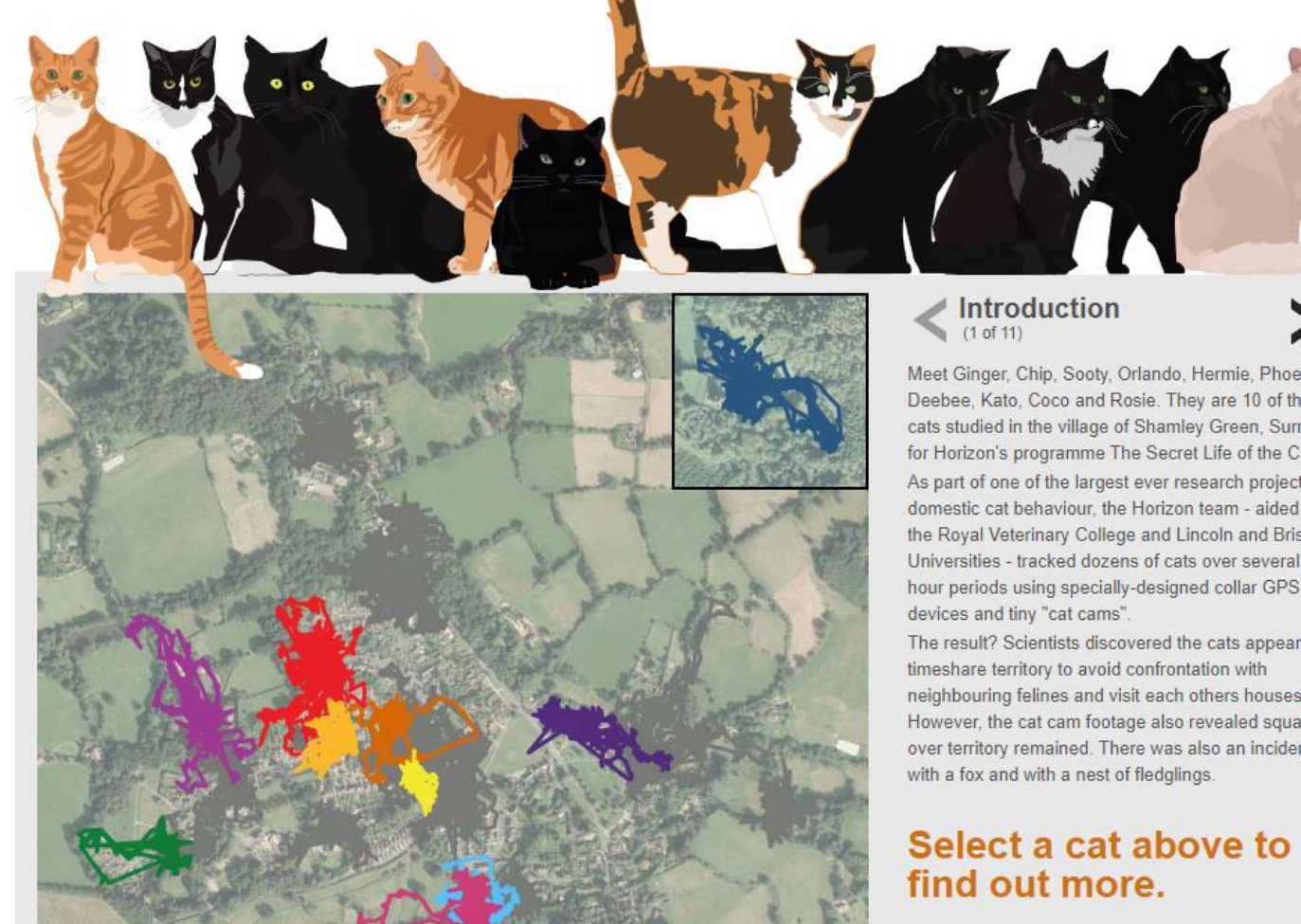




Brexit votes – Cambridge Intelligence



Interactive graphs



Allow for extra information – good

Depends on the user to extract information – less good. May be subject to technology moving on (e.g. Flash)

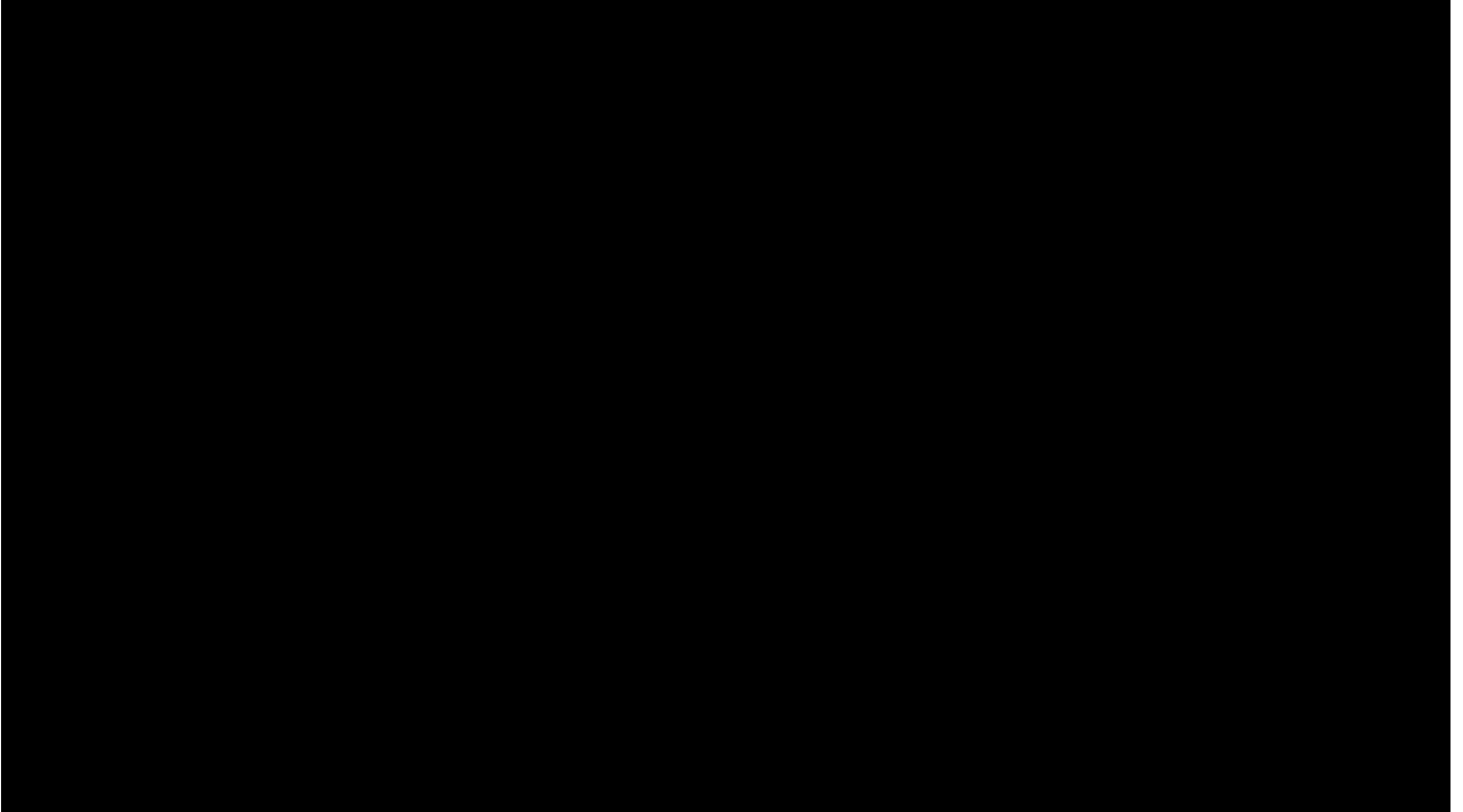
[Tree of life](#)

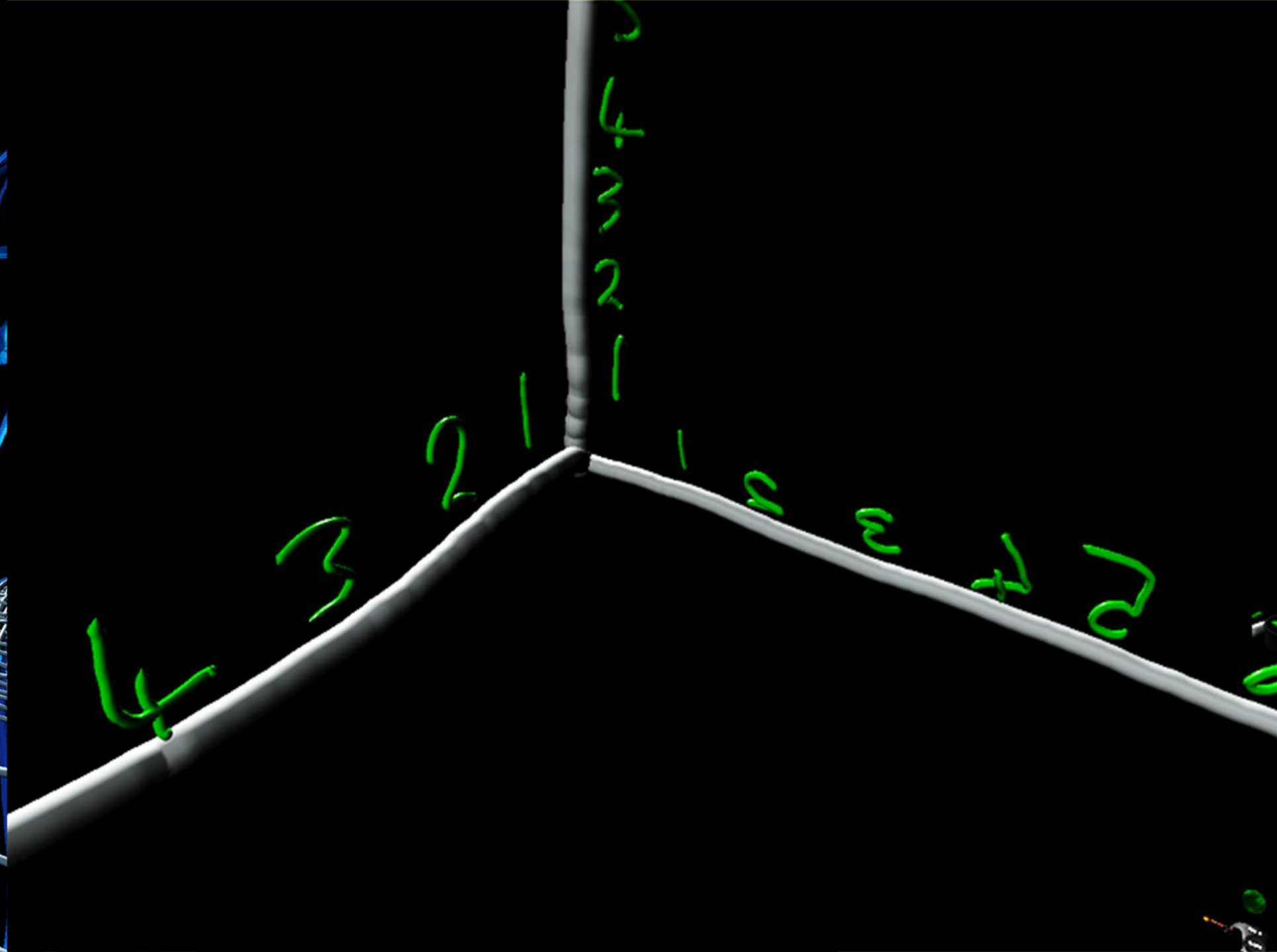
Motion graphs

<https://vimeo.com/33712288>

Allow a time dimension (usually to indicate progression)

Motion graphs – allow a time dimension





Understanding graphs

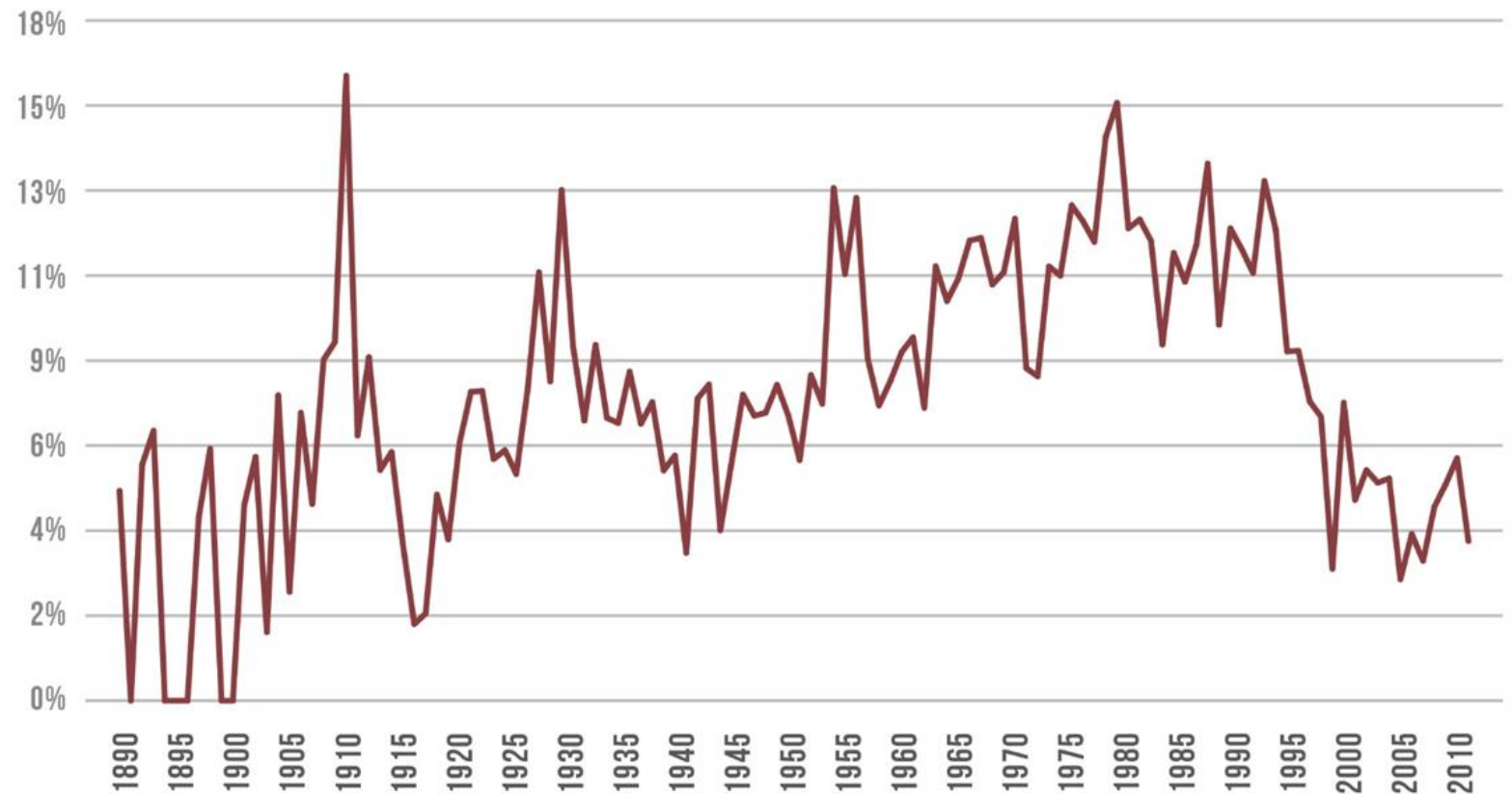
Start by looking at
how it is
constructed

Move on to what it
is trying to tell you

Finish by being
critical

LOVE IN SONG TITLES by year

idibon



Areas for comment

- Form:
- Title & footnotes

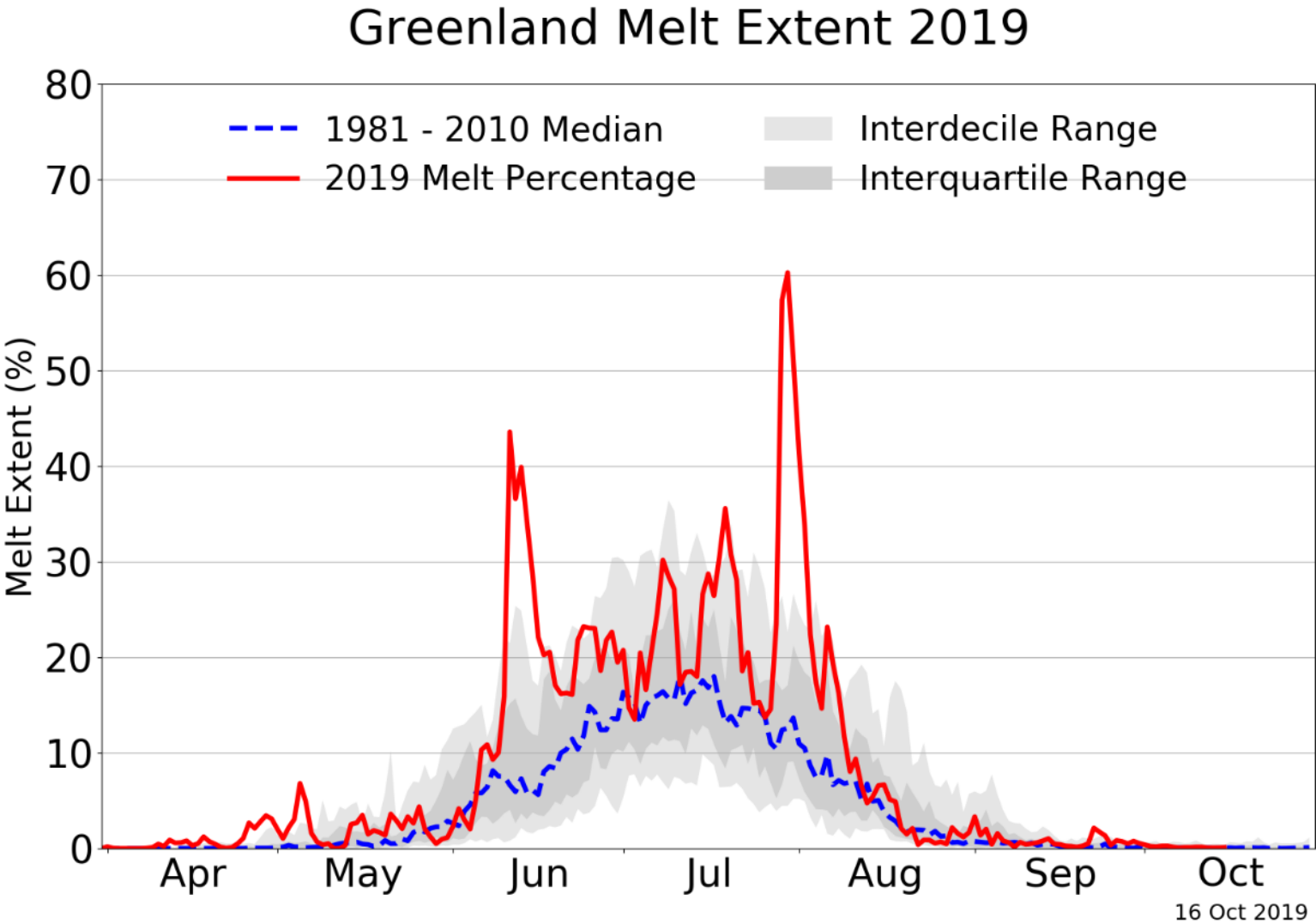
Axes labels/scale

Data Source

Colouration & symbols
- Message:
- Meaning

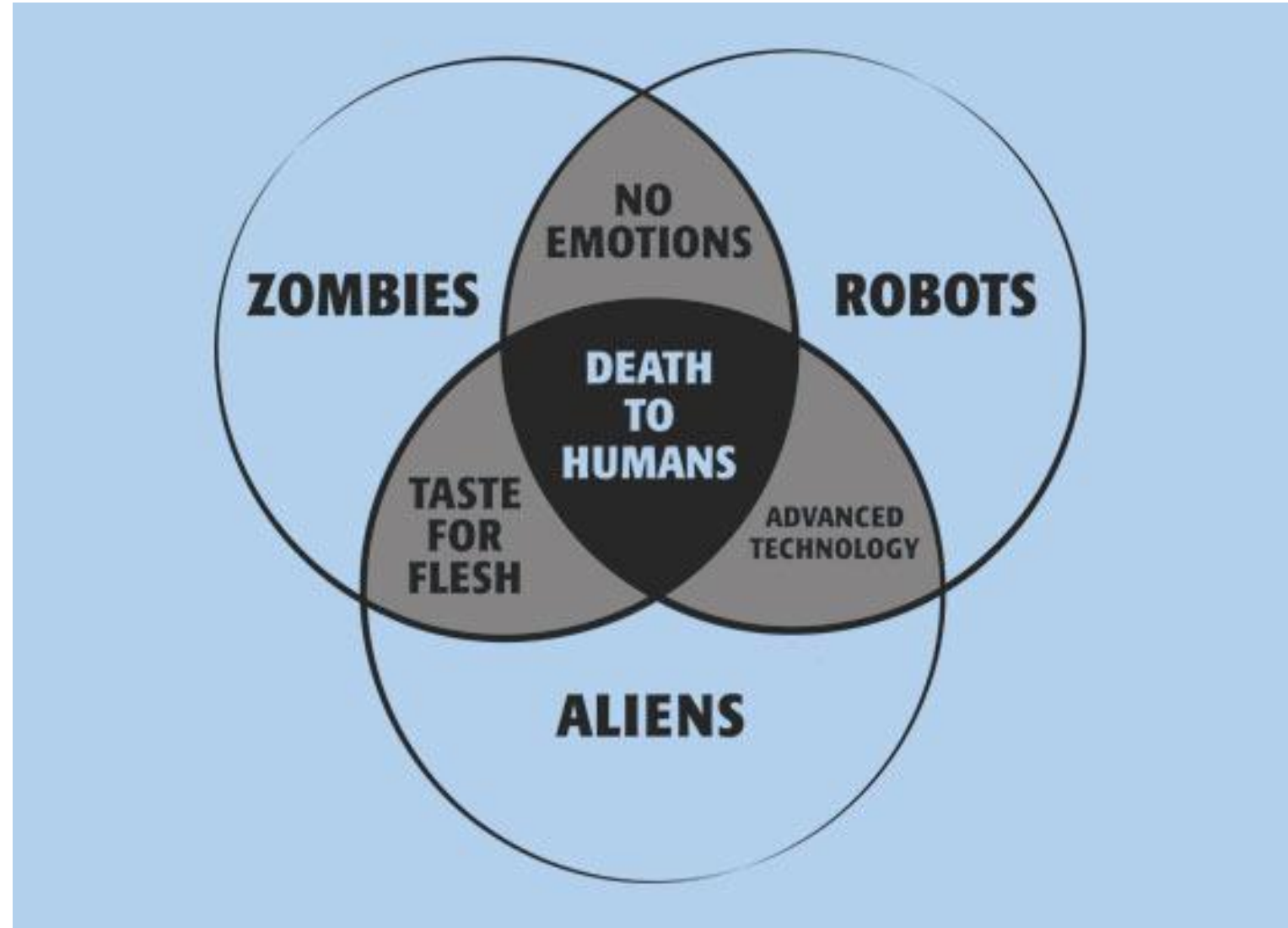
Implications
- Criticism:
- Missing data

Ease of understanding



Summary

- 1) Communicating complex data requires the use of images (graphs)
- 2) Understanding graphs is not always easy: take your time
- 3) Sometimes graphs are used to deliberately mislead: be critical



We will be using Minitab for this topic's seminars

Minitab is available for off-campus use here, in the Minitab license folder:

[Minitab sharepoint](#)

There are different versions for Mac and PC. There is an 'express' version for laptops too. Make sure that you go for the most recent version(19) and that you have downloaded the license file from this location too!